

Software Test Plan (STP)

Smart Complaint Management System

Field	Detail
Project	Smart Complaint Management System (SCMS)
Version	1.0
Authors	Team 8 — Aryaman D K, Vikas Gowda L, V S R Vinay Varma, Shridhar Umesh Golipalle
Date	02-10-2026
Status	Draft

1. Introduction

Purpose

This document defines the test plan for the Smart Complaint Management System (SCMS) v1.0. It states the objectives, scope, strategy, environment, schedule, resources and responsibilities for testing, and lists the test cases that verify every requirement baselined in the SRS.

Scope

Testing covers the complaint life cycle: registration and authentication, complaint lodging with evidence, routing and assignment, SLA escalation, status tracking, resolution and closure, notification, reporting and audit. It also covers the non-functional and security requirements of the SRS. Excluded: the internal behaviour of the e-mail and SMS gateways, browser and operating-system internals, and the PostgreSQL engine itself.

References

- SCMS Software Requirements Specification (SRS) v1.0, 06-09-2026 — source of all requirement IDs and acceptance criteria.
- SCMS Software Architecture Document (SAD) v1.0, 30-09-2026 — source of component boundaries, API definitions and the STRIDE threat model.
- OWASP Top 10 (2021) and the OWASP Application Security Verification Standard.
- Web Content Accessibility Guidelines (WCAG) 2.1, Level AA.
- Jira project T8 — backlog and defect tracking.

Definitions

Term	Meaning
SCMS	Smart Complaint Management System
STP / SRS / SAD	Software Test Plan / Software Requirements Specification / Software Architecture Document
RTM	Requirements Traceability Matrix
SLA	Service Level Agreement — maximum permitted resolution time for a complaint category
UAT	User Acceptance Testing

2. Test Items

The items under test are the components defined in SAD section 3.4:

- AuthModule — registration, OTP, login, roles, password reset, logout
- ComplaintModule — lodging, reference ID generation, evidence attachments, guest flow, drafts
- RoutingModule — auto-routing, assignment, priority, SLA escalation, duplicate detection
- TrackingModule — state machine, comments, resolution gating, reopen, rating
- NotificationModule and the Celery worker — e-mail and SMS dispatch with retry
- AdminModule — master data, user and role management
- ReportModule — dashboards, search and filtering, CSV and PDF export
- AuditModule — append-only audit trail and retention
- REST API layer — the endpoints defined in SAD section 4.3
- Web UI — the role-scoped browser interface

3. Features to be Tested

All 42 requirements of SRS v1.0 are covered: 28 functional (SCMS-F-001 to SCMS-F-028), 7 non-functional (SCMS-NF-001 to SCMS-NF-007) and 7 security (SCMS-SR-001 to SCMS-SR-007). Grouped by suite:

Suite	Requirements	Focus
Authentication	SCMS-F-001 to SCMS-F-005	Registration, OTP, login, session expiry, role enforcement, reset, logout
Lodging	SCMS-F-006 to SCMS-F-010	Field validation, reference ID uniqueness, attachment limits, guest flow, drafts
Routing	SCMS-F-011 to SCMS-F-015	Category mapping, assignment, priority, SLA escalation timing, duplicate flagging
Resolution	SCMS-F-016 to SCMS-F-020	State transitions, comments, resolution gating, reopen window, rating
Notification	SCMS-F-021, SCMS-F-022	Dispatch latency, channel coverage, retry and failure logging
Administration & reporting	SCMS-F-023 to SCMS-F-028	Master data effect, user lifecycle, dashboard scoping, export fidelity, audit, search
Performance	SCMS-NF-001, SCMS-NF-002, SCMS-NF-003	Response time, availability, scaling behaviour
Accessibility & maintainability	SCMS-NF-004 to SCMS-NF-006	WCAG 2.1 AA, coverage and lint, container portability
Security	SCMS-SR-001 to SCMS-SR-007, SCMS-NF-007	TLS, password storage, authorisation, OWASP scan, file upload, audit immutability, rate limiting, retention

4. Features Not to be Tested

- Internal behaviour of the SMTP e-mail service and the SMS gateway — third-party services, stubbed during testing.
- PostgreSQL, Redis, Django and Celery internals — assumed correct as released upstream.
- Browser and operating-system behaviour beyond the supported versions listed in SRS section 2.4.
- Payment processing, native mobile applications and external grievance-system integration — out of scope in SRS section 1.2.
- Load beyond 500 concurrent sessions — outside the stated scalability target.

5. Test Approach / Strategy

Levels

- Unit — per Django app, with pytest; the target is at least 70% line coverage (SCMS-NF-005).
- Integration — module to module and module to external stub: routing after lodging, notification after escalation, audit after every transition.
- System — end-to-end flows through the browser, matching sequence diagrams SD-01 and SD-02 of the SAD.
- Acceptance — each SRS acceptance criterion demonstrated against the deployed build.

Types

- Functional — the 28 functional requirements, including negative and boundary cases.
- Regression — the full automated suite re-run on every merge to the main branch.
- Performance — JMeter at 200 and 500 concurrent users.
- Usability and accessibility — Axe scan, keyboard-only navigation, 320 px rendering.
- Security — the seven security requirements plus an OWASP ZAP baseline scan.

Entry criteria

- A stable build is deployed to the test environment and passes the smoke suite.
- Master data is seeded: departments, categories, category-to-department mapping, SLA durations, priority matrix.
- Test accounts exist for all four roles, and e-mail and SMS gateway stubs are running.
- Test cases are written, reviewed and loaded into Jira.

Exit criteria

- 100% of planned test cases executed.
- No critical defect open, and no major defect open against a High-priority requirement.
- Every High-priority functional requirement passes its acceptance criterion.
- All seven security test cases pass and the ZAP scan shows no High or Medium finding.
- No non-functional requirement fails its stated measurement.
- Unit test line coverage at or above 70%.

5.1 Security Validation

- Password and OTP handling: storage as Argon2id hashes, never logged, never echoed to the UI (TC-Sec-02).
- TLS 1.2+ verification with HSTS, and HTTP-to-HTTPS redirection (TC-Sec-01).
- Authorisation bypass attempts: changing complaint IDs in URLs across users and departments (TC-Sec-03).
- Input fuzzing of complaint description, subject, comment and search fields with SQL and script payloads (TC-Sec-04).
- File upload abuse: disguised executables, oversized files, direct-URL execution attempts (TC-Sec-05).
- Audit immutability: UPDATE and DELETE attempted with the application database account (TC-Sec-06).
- Abuse resistance: submission and login rate limits, and CAPTCHA on the guest form (TC-Sec-07).
- Each test maps to a STRIDE category in SAD section 3.9.1, so threat coverage can be checked directly.

6. Test Environment

Element	Detail
Hardware	No special hardware. Test server: 2 vCPU, 4 GB RAM, 40 GB storage. Clients: desktop browser plus one Android and one iOS device for responsive and accessibility checks
Software	SCMS v1.0 on Ubuntu 22.04, Python 3.11, Django 5.x, PostgreSQL 14, Redis, Celery worker and Celery Beat, served by Unicorn behind Nginx. Browsers: Chrome, Firefox, Edge and Safari, latest two versions. Local SMTP sink and a mock SMS gateway with controllable failure modes
Tools	pytest + coverage (unit), Postman (API), Selenium (UI regression), JMeter (performance), OWASP ZAP (security), Axe (accessibility), flake8 (lint), Jira project T8 (test and defect tracking), GitHub Actions (CI)

Element	Detail
Test Data	Seeded departments and categories with known SLA values; test users for all four roles; sample complaints in every state; sample JPG, PNG and PDF attachments plus deliberately invalid files; a bulk script for the 10,000-complaint uniqueness check. No real complainant data is used

7. Test Schedule

Milestone	Date
Test plan baselined (this document)	02-10-2026
Test case design complete and reviewed	07-10-2026
Test environment and data ready	09-10-2026
Test execution starts (functional suites)	10-10-2026
Integration and system testing	14-10-2026 to 20-10-2026
Performance and security testing	21-10-2026 to 24-10-2026
Test execution ends; defects triaged	27-10-2026
User acceptance testing	28-10-2026 to 31-10-2026
Test summary report and sign-off	03-11-2026

Dates assume the implementation milestones hold. Any slip in build delivery shifts execution by the same amount; the plan is not compressed to absorb it, because that would mean dropping test cases.

8. Test Deliverables

- Test Plan (this document).
- Test cases (manual and automated) — 42 cases, one per requirement, held in Jira project T8 and linked to their requirement IDs.
- Automated test scripts — pytest unit and integration suites, Selenium UI regression suite, JMeter performance plan.
- Test data set and seeding script.
- Test execution logs and the CI run history.
- Defect reports in Jira, with severity and the requirement affected.
- Coverage report, OWASP ZAP scan report and Axe accessibility report.
- Test Summary Report with the final pass or fail position against the exit criteria.

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	V S R Vinay Varma	Prepare the plan, coordinate execution, triage defects, own the exit criteria and the summary report
Test Engineer	Shridhar Umesh Golipalle	Design and execute test cases, build the environment and test data, log defects, run the performance and security tools
Developer	Vikas Gowda L	Support defect triage and fixes, maintain unit tests, deliver stable builds to the test environment
Product Owner	Aryaman D K	Approve test results, run UAT against the acceptance criteria, sign off release readiness

10. Risks and Mitigation

Risk	Mitigation
Build delivered late, compressing the execution	Request early smoke builds; execute High-priority suites first so the most important

Risk	Mitigation
window	coverage is complete even if time runs out
SLA escalation cases are slow to test because they depend on elapsed time	Make SLA durations configurable and seed complaints with back-dated deadlines rather than waiting in real time
Malware scanning for TC-Sec-05 may not be available in the student environment	Raised as OI-05 in the SAD. If ClamAV cannot run in the container, record the gap explicitly in the summary report rather than marking the test passed
Gateway stubs behave differently from the real SMS provider	Test against the real provider in sandbox mode at least once before sign-off
Test environment unavailable or shared with development	Separate database and container stack for testing; restore from the seeding script when corrupted
Four part-time testers, with execution overlapping other coursework	Automate the regression suite early; keep manual effort for exploratory and UAT work
Defects found late in security testing need architectural change	Run the ZAP baseline scan from the first integration build rather than only in the security phase

11. Assumptions & Dependencies

- The test environment mirrors the deployment configuration in SAD section 3.6.
- Master data and test accounts are seeded before execution begins.
- E-mail and SMS gateway stubs are available, with controllable failure modes.
- Requirements remain as baselined in SRS v1.0; any change is re-baselined and the affected test cases revised.
- Developers deliver builds that pass the smoke suite before handover to test.
- Jira project T8 remains available for test and defect tracking.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours.
- The build blocks more than 30% of planned test cases.
- A critical defect makes further results unreliable — for example an authorisation failure that exposes other users' complaints.
- Test data is corrupted such that results cannot be reproduced.

Resume testing when:

- Blocking defects are resolved and verified on a new build.
- The environment is stable and the smoke suite passes.
- Test data has been reseeded and verified.
- Test cases affected by the suspension have been re-run from the start.

13. Test Case Management & Traceability

The RTM ensures that every requirement in SRS v1.0 maps to at least one test case. Test cases are held in Jira project T8, each linked to the issue carrying its SRS requirement ID. The detailed test cases themselves are a separate deliverable (section 8). The mapping is:

Requirement	Test case	Requirement	Test case	Requirement	Test case
SCMS-F-001	TC-Auth-01	SCMS-F-002	TC-Auth-02	SCMS-F-003	TC-Auth-03
SCMS-F-004	TC-Auth-04	SCMS-F-005	TC-Auth-05	SCMS-F-006	TC-Lodge-01
SCMS-F-007	TC-Lodge-02	SCMS-F-008	TC-Lodge-03	SCMS-F-009	TC-Lodge-04
SCMS-F-010	TC-Lodge-05	SCMS-F-011	TC-Route-01	SCMS-F-012	TC-Route-02
SCMS-F-013	TC-Route-03	SCMS-F-014	TC-Route-04	SCMS-F-015	TC-Route-05
SCMS-F-016	TC-Track-01	SCMS-F-017	TC-Track-02	SCMS-F-018	TC-Res-01
SCMS-F-019	TC-Res-02	SCMS-F-020	TC-Res-03	SCMS-F-021	TC-Notif-01
SCMS-F-022	TC-Notif-02	SCMS-F-023	TC-Admin-01	SCMS-F-024	TC-Admin-02
SCMS-F-025	TC-Rep-01	SCMS-F-026	TC-Rep-02	SCMS-F-028	TC-Rep-03
SCMS-F-027	TC-Audit-01	SCMS-NF-007	TC-Audit-02	SCMS-NF-001	TC-Perf-01
SCMS-NF-003	TC-Perf-02	SCMS-NF-002	TC-Ops-01	SCMS-NF-004	TC-UX-01
SCMS-NF-005	TC-Maint-01	SCMS-NF-006	TC-Port-01	SCMS-SR-001	TC-Sec-01
SCMS-SR-002	TC-Sec-02	SCMS-SR-003	TC-Sec-03	SCMS-SR-004	TC-Sec-04
SCMS-SR-005	TC-Sec-05	SCMS-SR-006	TC-Sec-06	SCMS-SR-007	TC-Sec-07

Coverage: 28 functional + 7 non-functional + 7 security = 42 requirements, each mapped to a test case. No requirement is untested and no test case is orphaned.

14. Test Metrics & Reporting

Metrics collected

Metric	Definition and target
Test case execution rate	Executed cases divided by planned cases; target 100% before exit
Pass rate	Passed divided by executed; tracked per suite so a weak area is visible
Requirement coverage	Requirements with at least one executed case; target 100% of the 42
Defect density	Defects per module, to show where rework is concentrated
Defect aging	Days a defect stays open by severity; critical defects escalate after 2 days
Defect leakage	Defects found in UAT that functional testing should have caught
Automated coverage	Line coverage from pytest; target at or above 70% (SCMS-NF-005)

Reports

- Daily execution status during the execution window: cases run, passed, failed, blocked.
- Weekly defect summary by severity and module, reviewed by the whole team.
- Final Test Summary Report: results against each exit criterion, open defects with severity, requirement coverage, and an explicit statement of any requirement that could not be fully tested.

15. Approvals

Role	Name	Signature / Date
QA Lead	V S R Vinay Varma	
Dev Lead	Vikas Gowda L	
Product Owner	Aryaman D K	